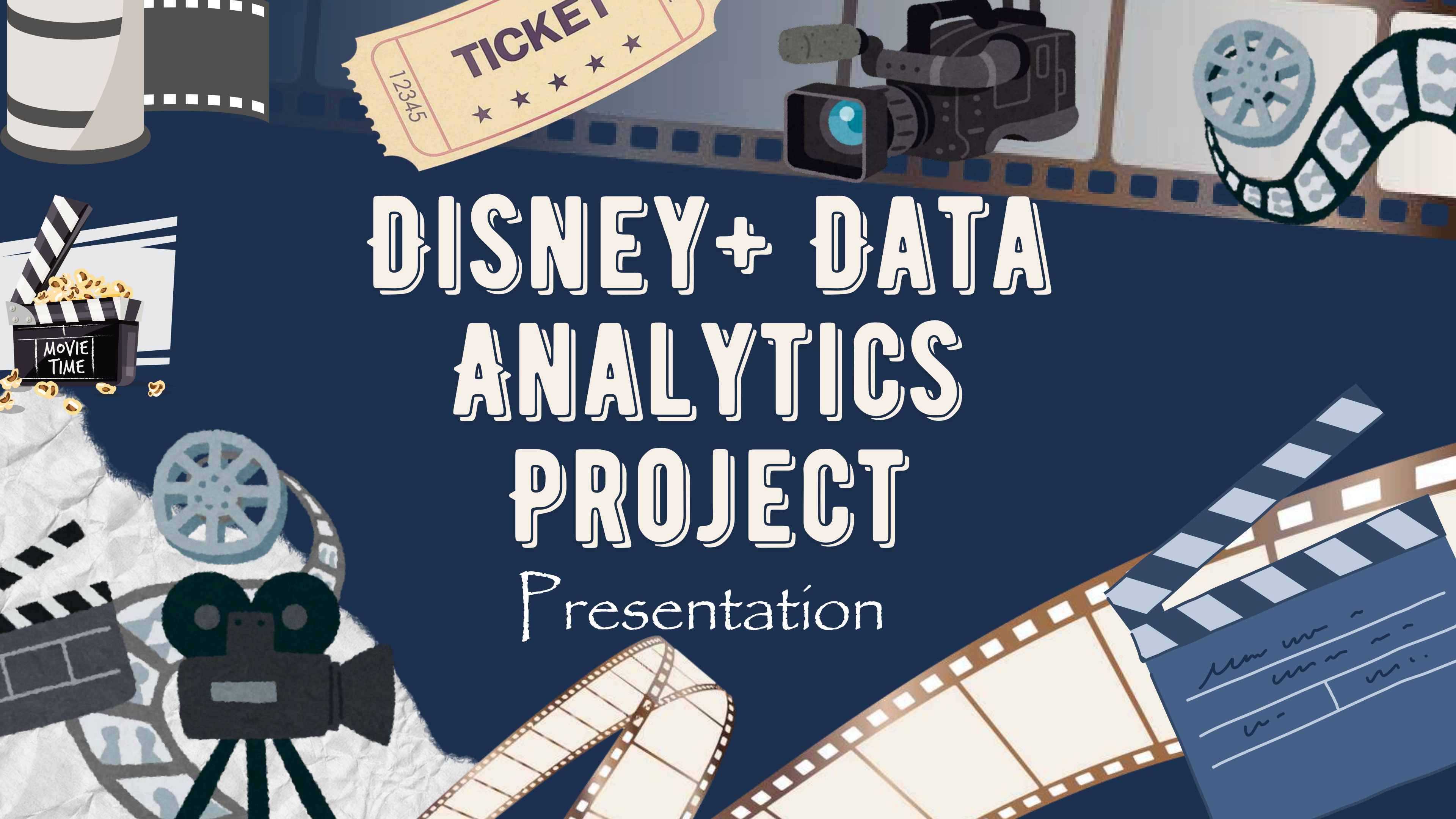
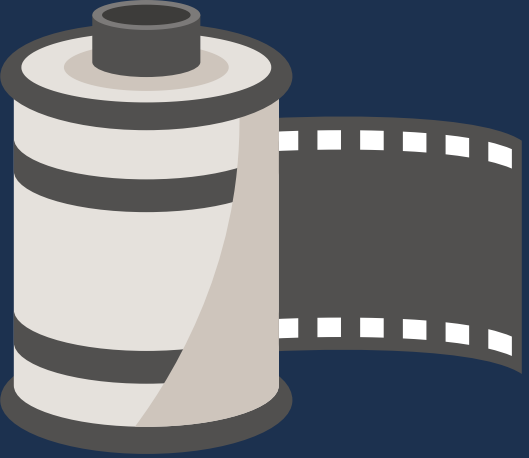


A collage of movie-related items on a dark blue background. At the top left is a film canister and a strip of film. Next to it is a yellow movie ticket with the word 'TICKET' and the number '12345'. To the right is a professional video camera. Further right is a film reel. In the bottom left, there's a clapperboard with 'MOVIE TIME' and popcorn, a camera on a tripod, and another film reel. At the bottom right, there's a large clapperboard with some handwritten text and a strip of film.

DISNEY+ DATA ANALYTICS PROJECT Presentation



OUR TEAM

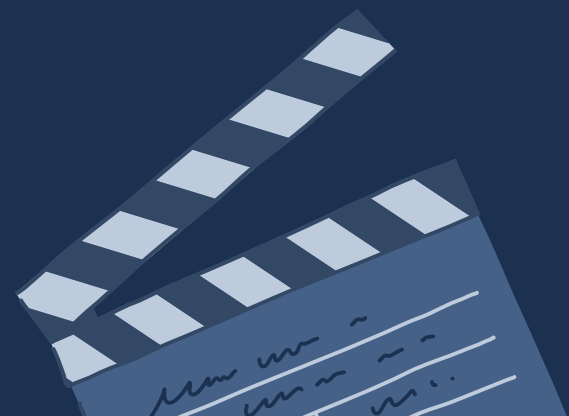
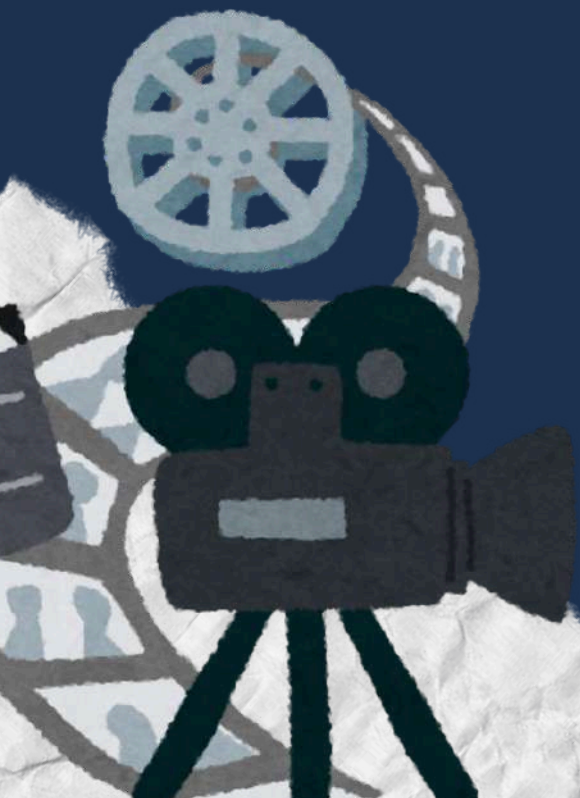
Shahd
Mohamed
Khedr

Shahd
Hany
Abdelrazek
Saeed

Hagar
Khalaf
Mostafa

Arwa
Sameh
Arafat

Samah
Samy
Elsayed
Ahmed





PROJECT OVERVIEW & OBJECTIVES

2. Business Problem

Streaming platforms require data-driven insights to optimize content acquisition, understand regional distribution preferences, and enhance user engagement.

1. Objective

Analyze the Disney+ catalog to uncover content distribution, release trends, target audience demographics, and genre preferences.

3. Scope

End-to-end data pipeline covering raw extraction, Python preprocessing, SQL querying, Excel reporting, and Power BI visualization.



TARGET AUDIENCE & STAKEHOLDERS

- 🎬 Content Acquisition Team

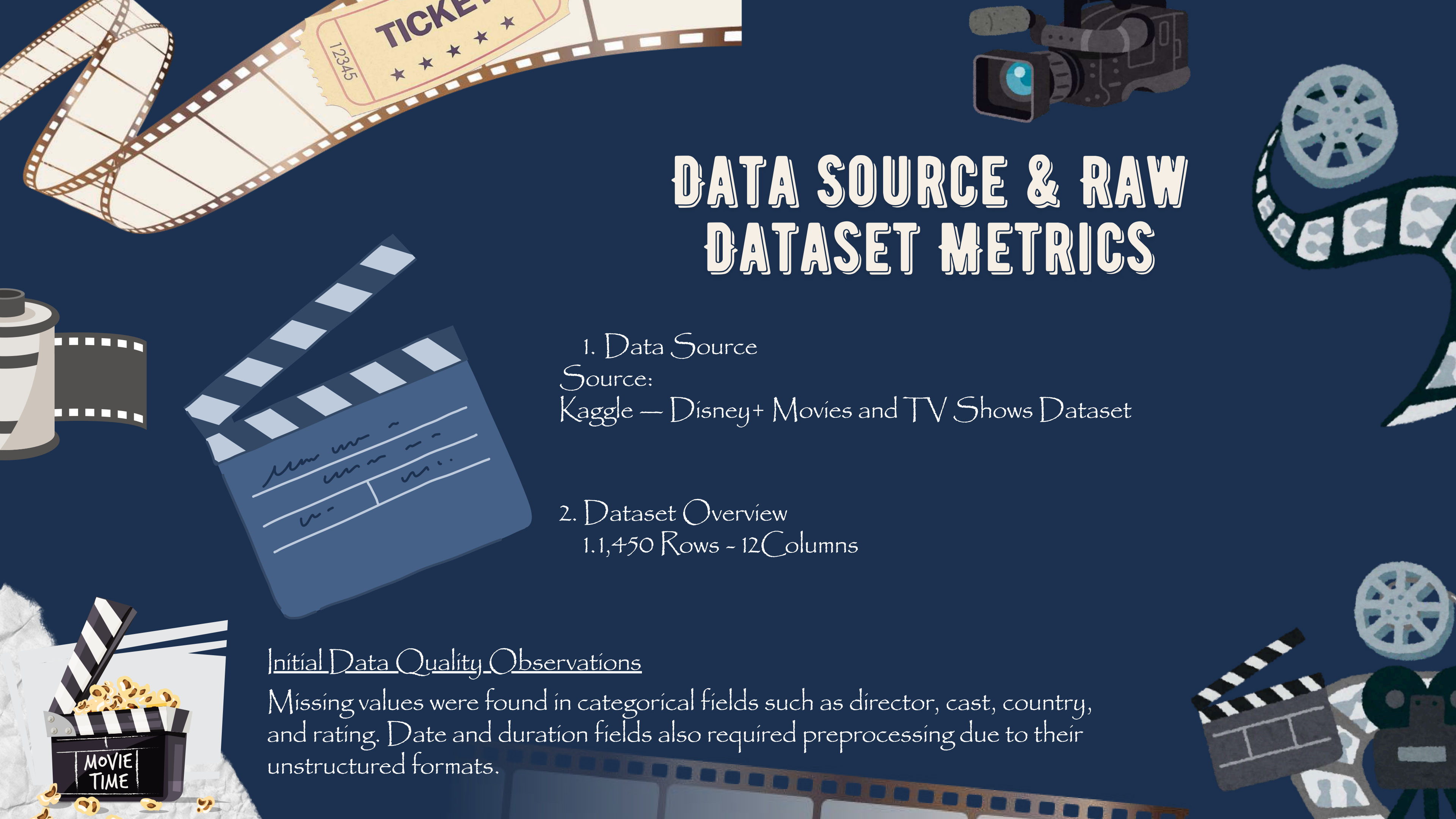
Identify gaps in genres, movie-to-TV show ratios, and popular content durations

- 📊 Executive Management

Support high-level strategic decisions regarding content portfolio expansion and platform growth.

- 📣 Marketing & Promotions Team

Tailor promotional campaigns based on release timing, regional popularity, and demographic ratings.



DATA SOURCE & RAW DATASET METRICS

1. Data Source

Source:

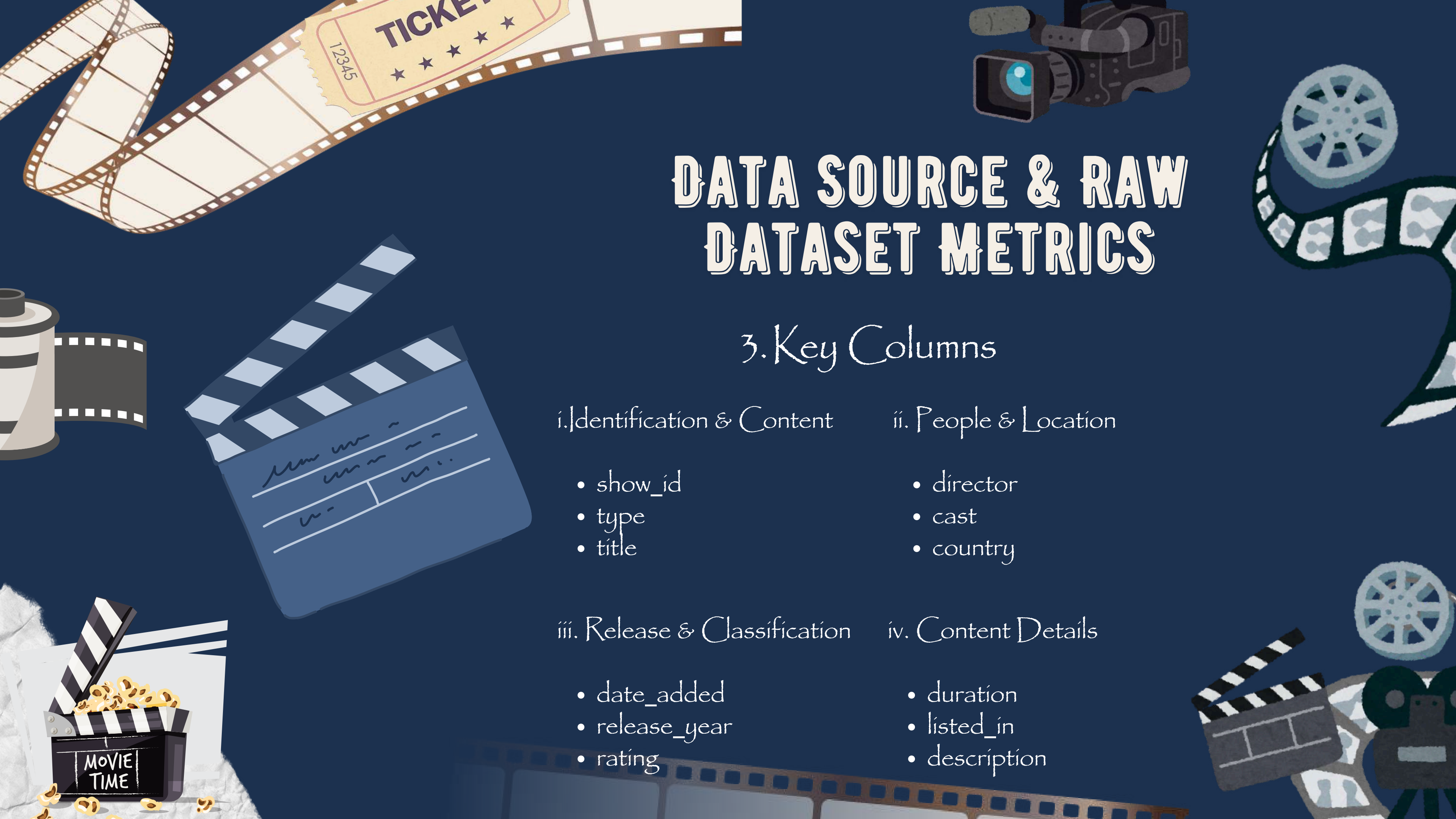
Kaggle — Disney+ Movies and TV Shows Dataset

2. Dataset Overview

1,1,450 Rows ~ 12 Columns

Initial Data Quality Observations

Missing values were found in categorical fields such as director, cast, country, and rating. Date and duration fields also required preprocessing due to their unstructured formats.



DATA SOURCE & RAW DATASET METRICS

3. Key Columns

i. Identification & Content

- show_id
- type
- title

ii. People & Location

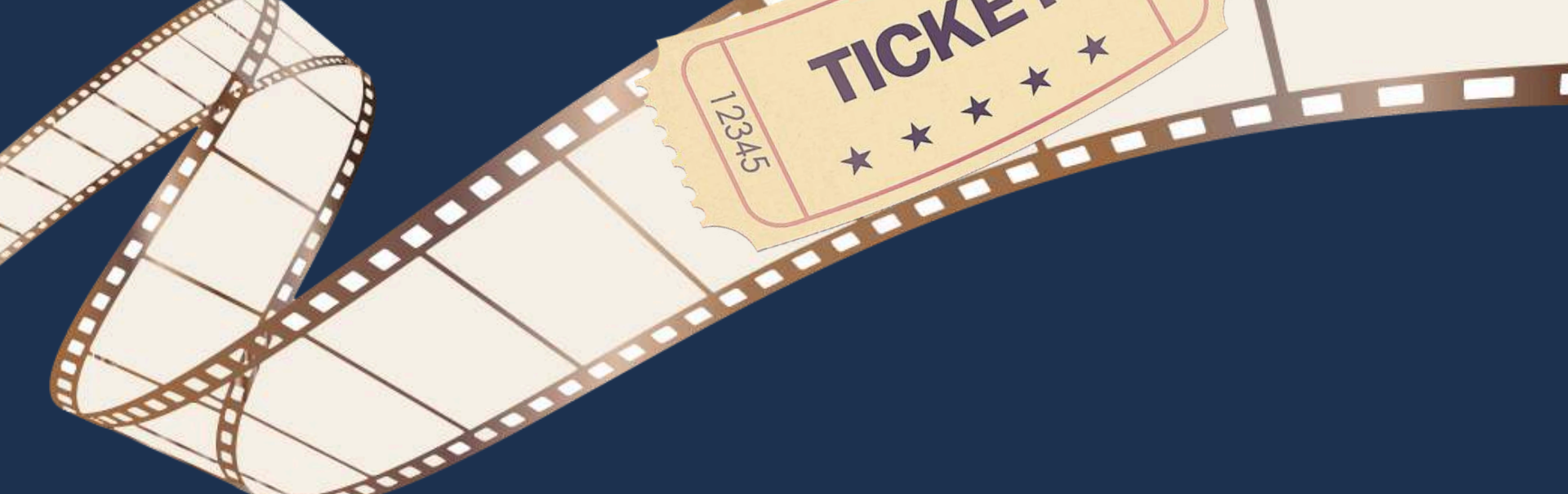
- director
- cast
- country

iii. Release & Classification

- date_added
- release_year
- rating

iv. Content Details

- duration
- listed_in
- description



1. Fill missing categorical variables with meaningful labels

```
df_clean['director'] = df_clean['director'].fillna('Unknown Director')
df_clean['cast'] = df_clean['cast'].fillna('Unknown Cast')
df_clean['country'] = df_clean['country'].fillna('Unknown Country')
```



1. Categorical Imputation

PYTHON – DATA CLEANING



Missing Values

Replaced missing values in director, cast, and country with:

Unknown Director

Unknown Cast

Unknown Country





```
# 2. Impute missing rating with Mode
mode_rating = df_clean['rating'].mode()[0]
df_clean['rating'] = df_clean['rating'].fillna(mode_rating)
print(f"Imputed missing ratings with Mode: '{mode_rating}'")
```

Imputed missing ratings with Mode: 'TV-G'

2. Rating Imputation

PYTHON – DATA CLEANING

Filled missing rating values using the statistical Mode:
TV-G.



PYTHON – DATA CLEANING

3. Date Filtering

Date Added

Removed a minimal number of rows with missing date_added values (<0.2%) where dates could not be reliably inferred.

```
# 3. Drop missing date_added rows (only 3 missing rows)
df_clean = df_clean.dropna(subset=['date_added']).reset_index(drop=True)
```

PYTHON – DATA CLEANING

1. Whitespace Stripping

Text Cleaning :

Removed leading and trailing whitespaces from all string and text columns to ensure clean joins and accurate groupings.

```
# 1. Trim whitespace from all string columns
str_cols = df_clean.select_dtypes(include=['object', 'string']).columns
for col in str_cols:
    df_clean[col] = df_clean[col].astype(str).str.strip()
```

PYTHON – DATA CLEANING

2. Duplicate Check

Data Integrity :

Verified record uniqueness across title and type.

Result: 0 duplicate records found

```
# 2. Check for duplicate records based on title and content type
duplicate_rows = df_clean.duplicated(subset=['title', 'type']).sum()
print(f"Total duplicate records found: {duplicate_rows}")
```

Total duplicate records found: 0

```
# Drop duplicates if any exist
if duplicate_rows > 0:
    df_clean = df_clean.drop_duplicates(subset=['title', 'type']).reset_index(drop=True)
    print("Duplicates removed successfully.")
else:
    print("No duplicates detected. Dataset is clean.")
```

No duplicates detected. Dataset is clean.

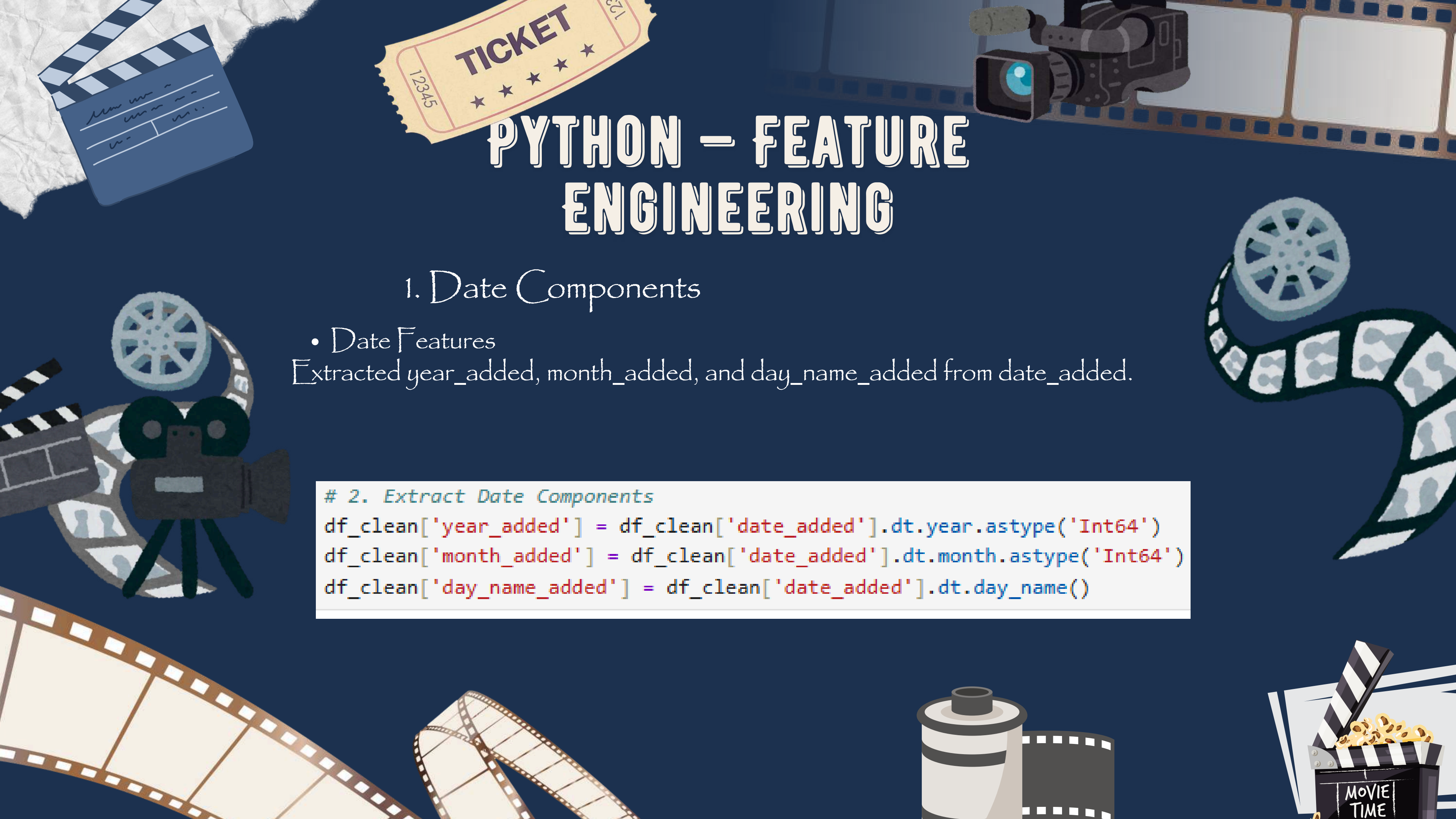
PYTHON – DATA CLEANING

3. Data Type Casting

Data Types :

Standardized column data types to prepare the dataset for database loading and numerical analysis.

```
# 1. Convert date_added to proper datetime format  
df_clean['date_added'] = pd.to_datetime(df_clean['date_added'].str.strip(), errors='coerce')
```



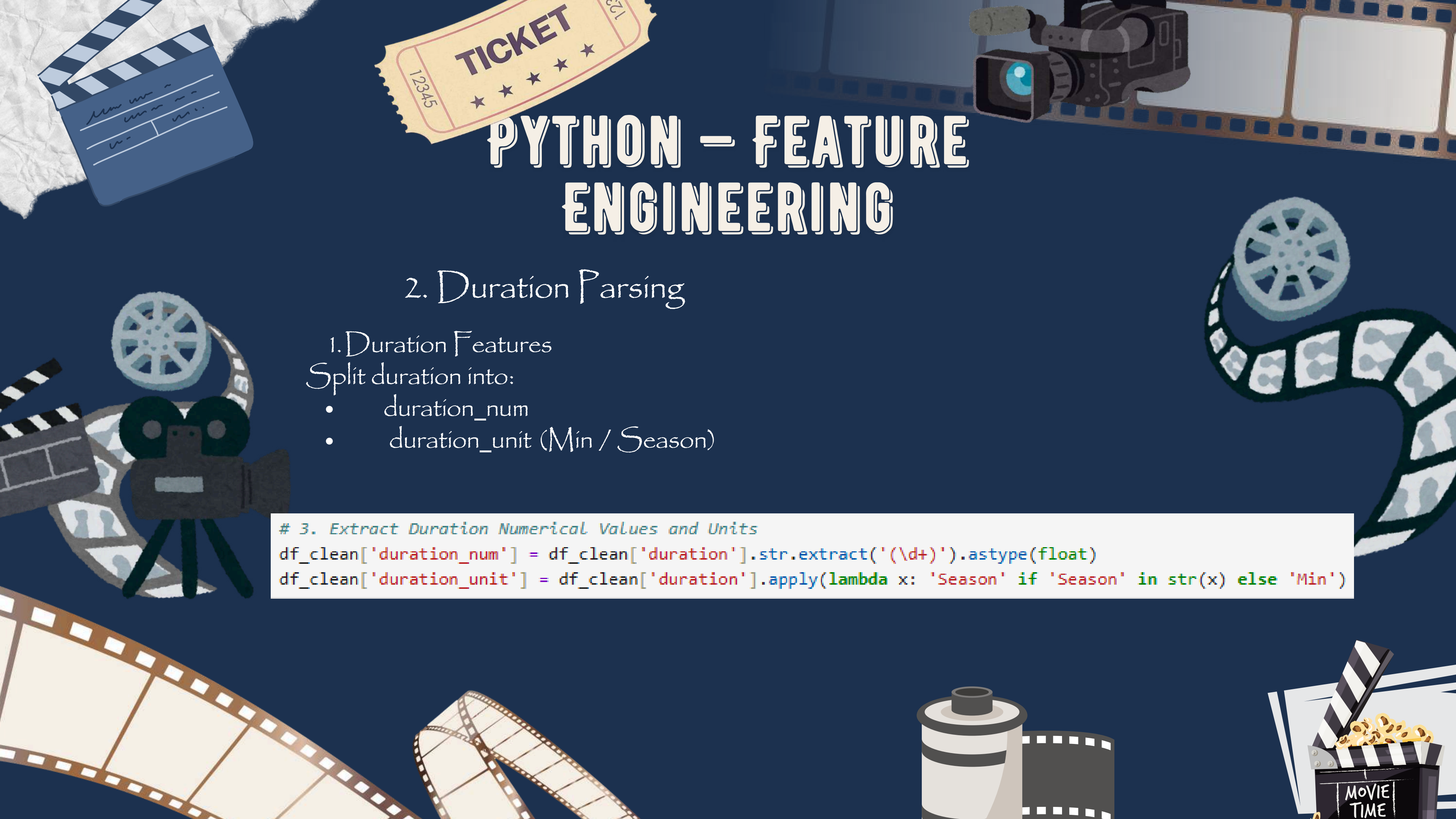
PYTHON – FEATURE ENGINEERING

1. Date Components

- Date Features
Extracted year_added, month_added, and day_name_added from date_added.

```
# 2. Extract Date Components
```

```
df_clean['year_added'] = df_clean['date_added'].dt.year.astype('Int64')  
df_clean['month_added'] = df_clean['date_added'].dt.month.astype('Int64')  
df_clean['day_name_added'] = df_clean['date_added'].dt.day_name()
```

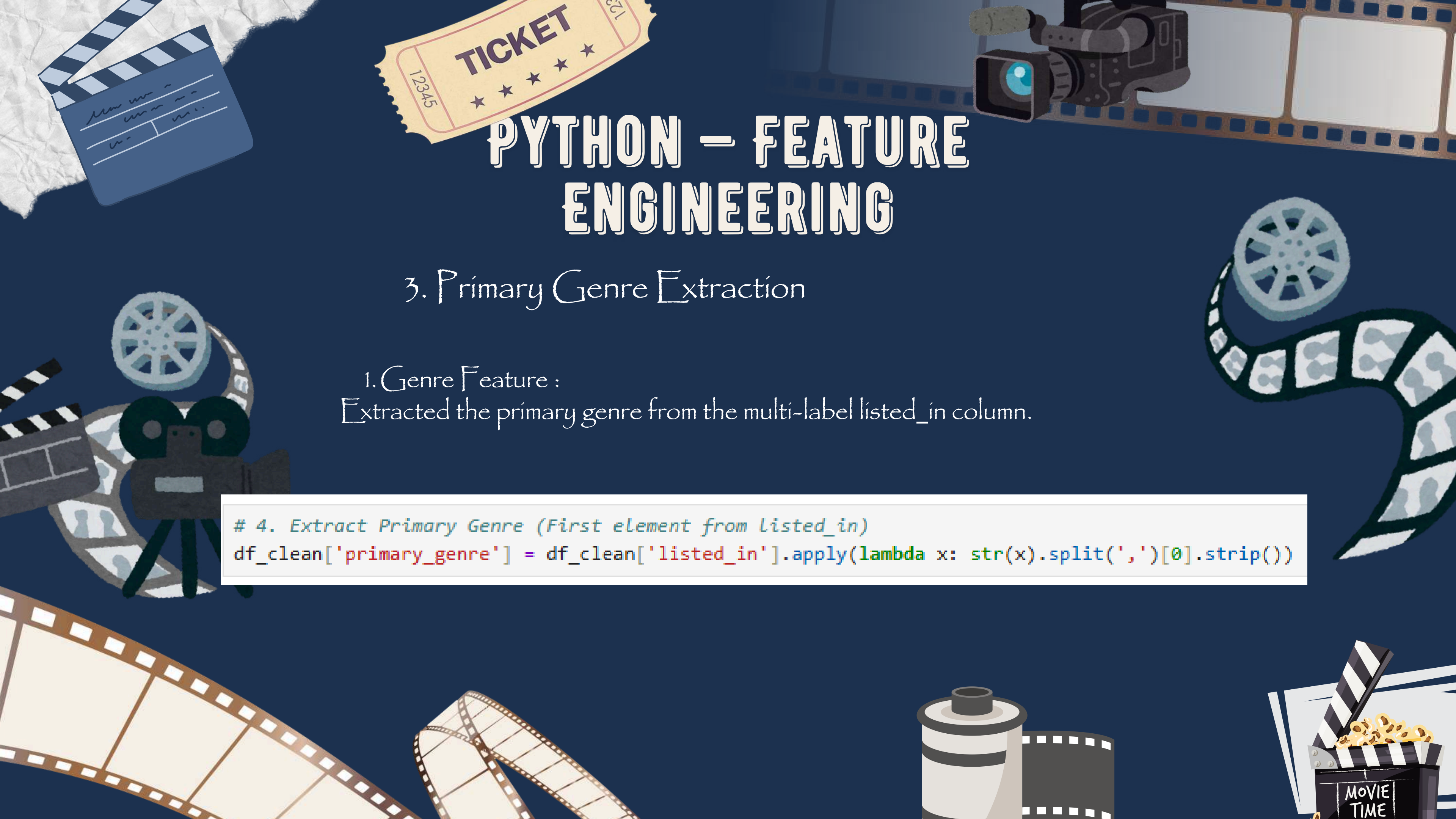


PYTHON – FEATURE ENGINEERING

2. Duration Parsing

1. Duration Features
- Split duration into:
- duration_num
 - duration_unit (Min / Season)

```
# 3. Extract Duration Numerical Values and Units
df_clean['duration_num'] = df_clean['duration'].str.extract('(\d+)').astype(float)
df_clean['duration_unit'] = df_clean['duration'].apply(lambda x: 'Season' if 'Season' in str(x) else 'Min')
```

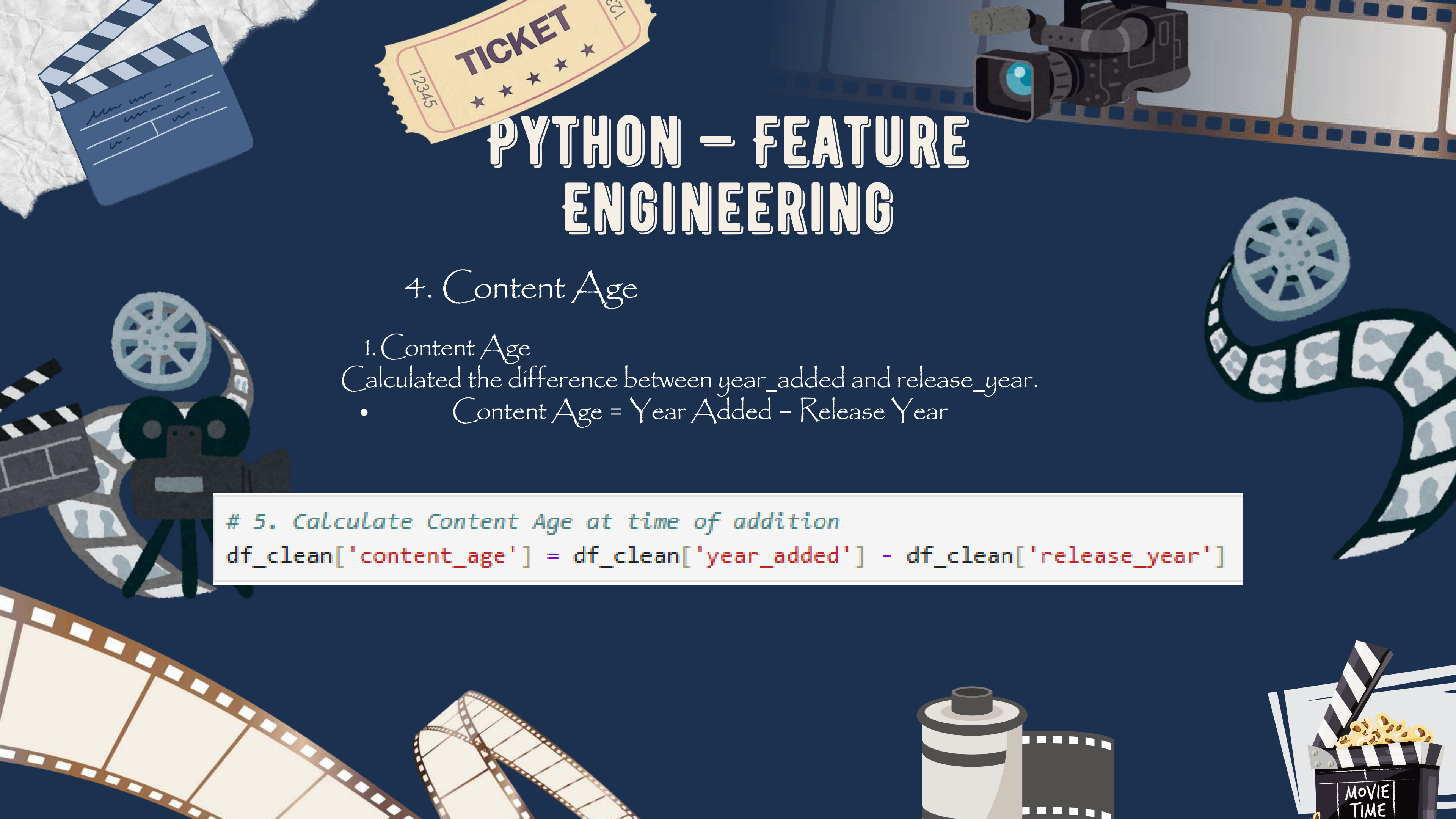


PYTHON – FEATURE ENGINEERING

3. Primary Genre Extraction

1. Genre Feature :
Extracted the primary genre from the multi-label listed_in column.

```
# 4. Extract Primary Genre (First element from listed_in)  
df_clean['primary_genre'] = df_clean['listed_in'].apply(lambda x: str(x).split(',')[0].strip())
```



PYTHON – FEATURE ENGINEERING

4. Content Age

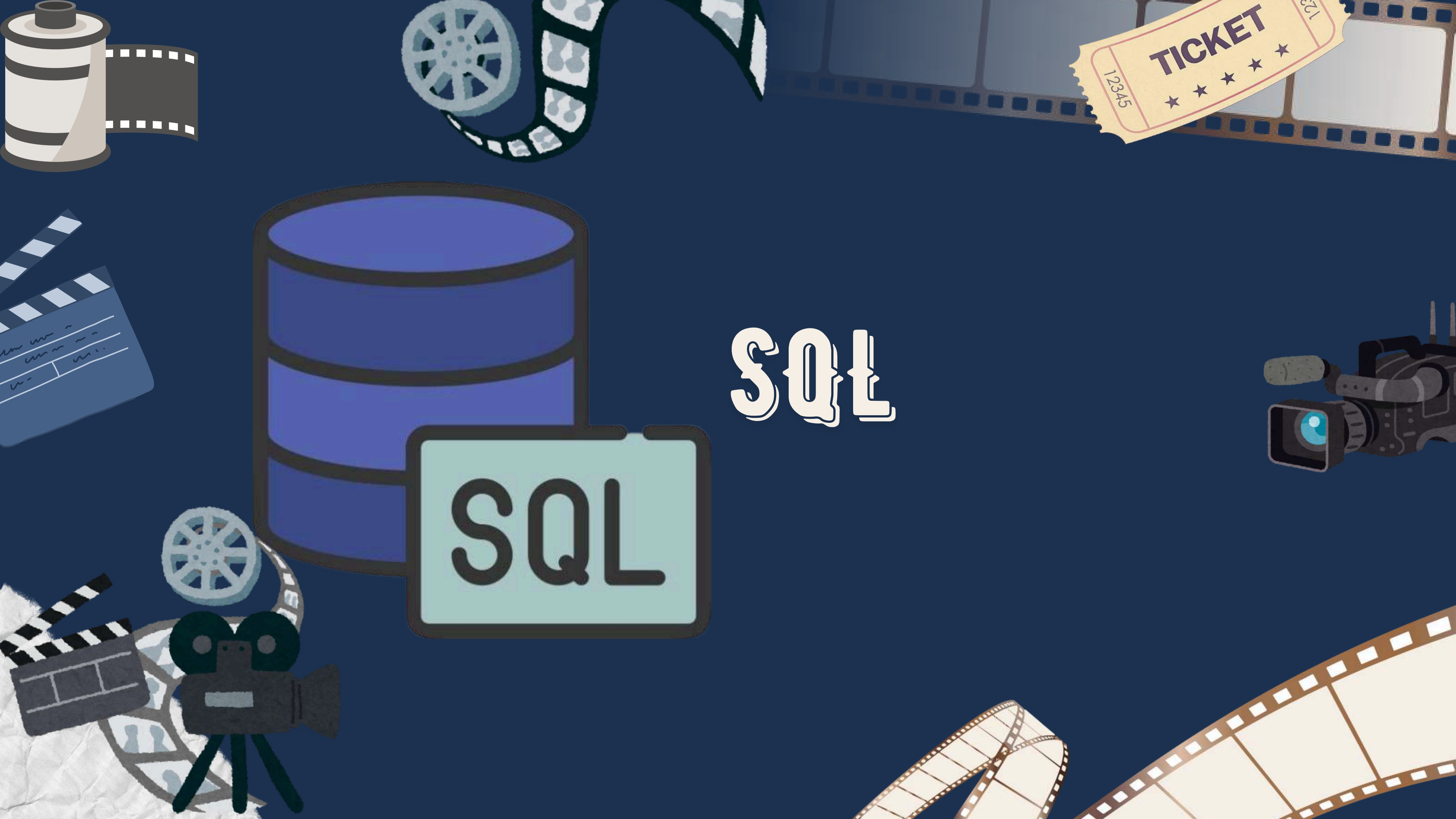
1. Content Age

Calculated the difference between `year_added` and `release_year`.

- $\text{Content Age} = \text{Year Added} - \text{Release Year}$

```
# 5. Calculate Content Age at time of addition
```

```
df_clean['content_age'] = df_clean['year_added'] - df_clean['release_year']
```



SQL

SQL



SQL ANALYSIS 1 – TIME TRENDS

Key Questions

- What is the total number of titles?
- How are Movies and TV Shows distributed?
- What is the average release year?
- Which month has the highest number of additions?

Key Insight

Content additions peak toward the final quarter of the year, with November showing the highest addition volume.



SQL ANALYSIS 2

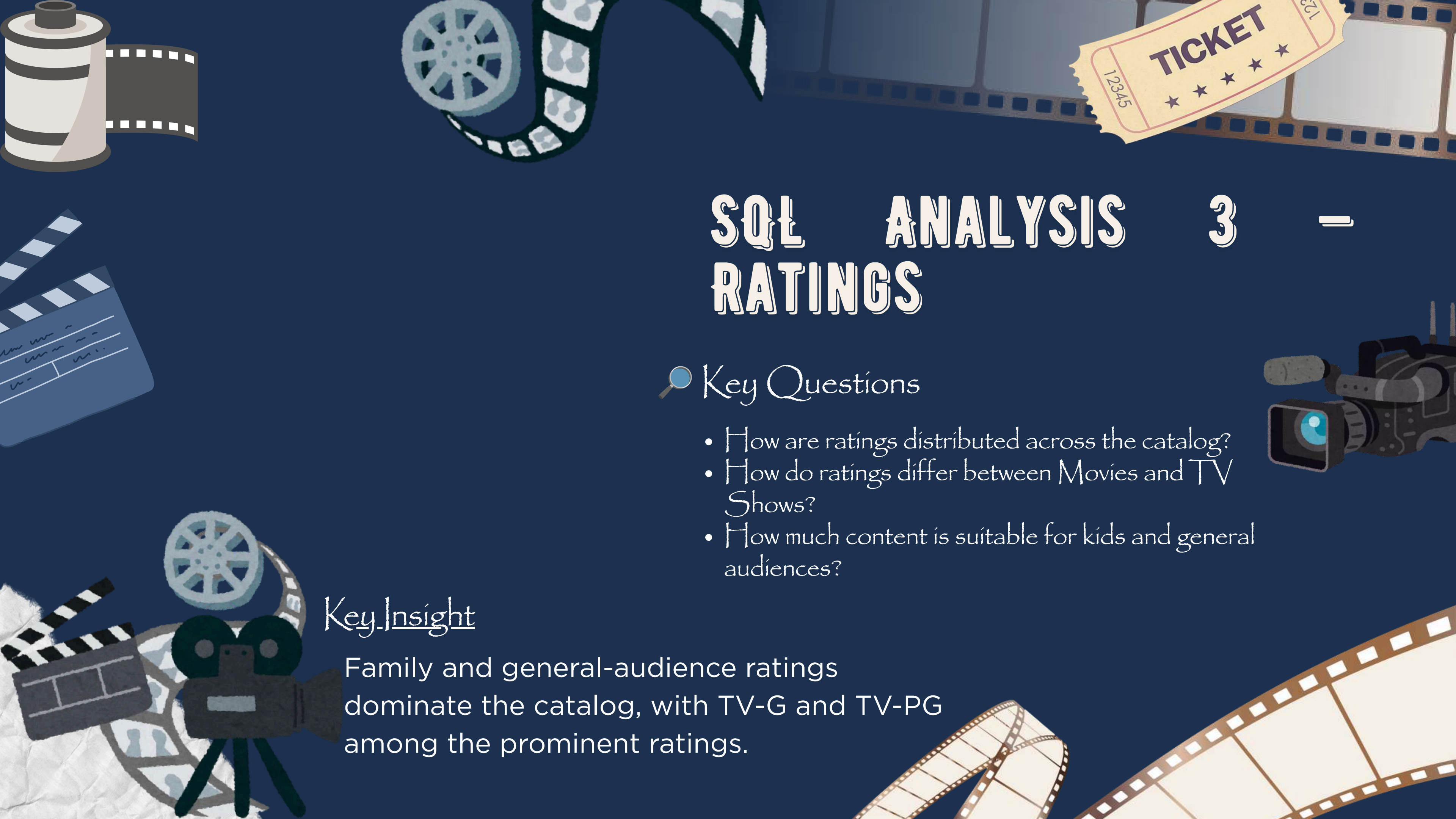
GENRES & DURATION

Key Questions

- What are the most popular primary genres?
- What is the average movie duration?
- How are TV shows distributed by number of seasons?

Key Insight

Animation, Comedy, and Family represent major content categories across the Disney+ catalog, covering both feature-length movies and shorter-form content.



SQL ANALYSIS 3

RATINGS

Key Questions

- How are ratings distributed across the catalog?
- How do ratings differ between Movies and TV Shows?
- How much content is suitable for kids and general audiences?

Key Insight

Family and general-audience ratings dominate the catalog, with TV-G and TV-PG among the prominent ratings.



SQL ANALYSIS 4 – COUNTRY, DIRECTOR & CAST

Key Questions

- Which countries produce the most content?
- Which directors appear most frequently?
- Which actors have the most appearances?
- How is content distributed across regions?

Key Insight

The United States leads content production by a large margin, while some older titles contain unknown or incomplete director and cast information.

A dark blue background decorated with various movie-related icons. In the top left is a professional video camera. To its right is a film canister and a strip of film. Further right is a clapperboard with 'MOVIE TIME' written on it, surrounded by popcorn. Next to that is a yellow movie ticket with 'TICKET' and '12345' printed on it. On the right side, there's a film reel and another strip of film. At the bottom, there's a film strip on the left, a clapperboard in the center, a movie camera on a tripod, and a crumpled white paper on the right.

EXCEL DASHBOARD – COUNTRY, DIRECTOR & CAST

-  Tool Selection

Excel was selected for flexible spreadsheet-based analysis and reporting.

-  Implementation

Built using Pivot Tables and interactive Slicers to filter the analysis by country and content type.

-  Purpose

Provides an interactive view of content distribution across countries, directors, and cast.



MOVIES ANALYTICS DASHBOARD

Most Frequent Actor

Jim Cummings (24)

Avg Titles per Director

2.36

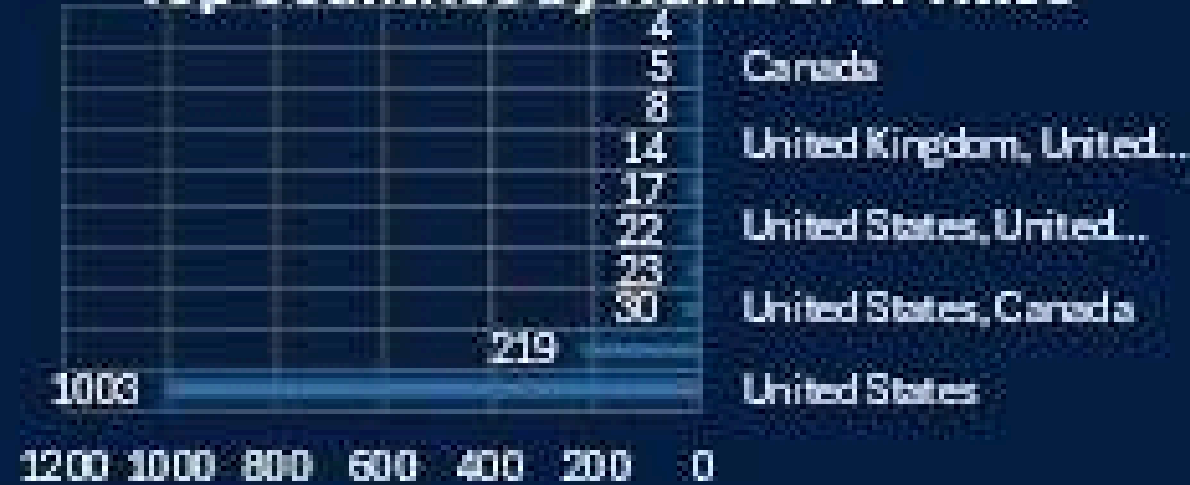
Unique Directors

674

Unique Countries

89

Top Countries by Number of Titles



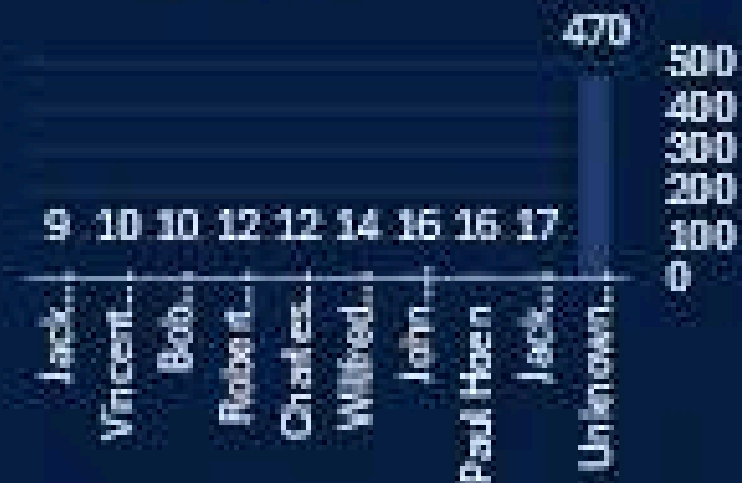
US Production %

81.60%

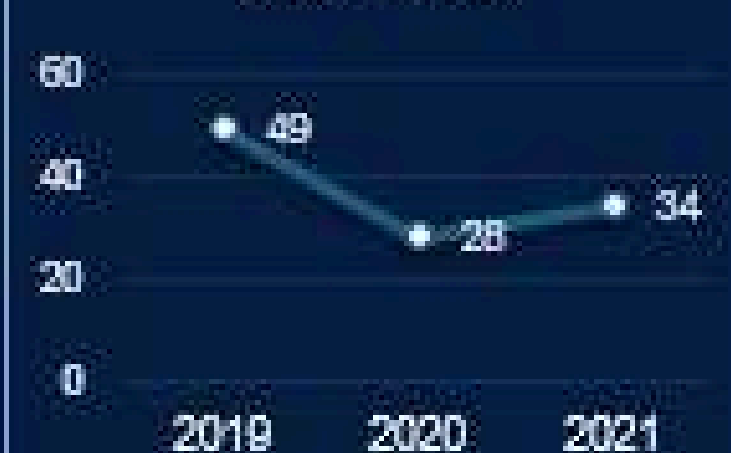
Unknown Country %

15.10%

Top 10 Directors



Countries Diversity Over Years



director

Alan Zas...

Andrew ...

Andy Byatt

Angus M...

Art Steve...

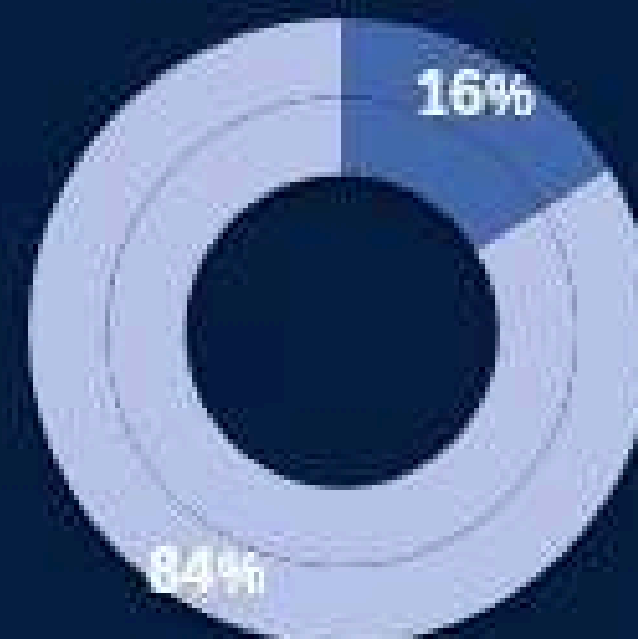
Beverly ...

Bill Robe...

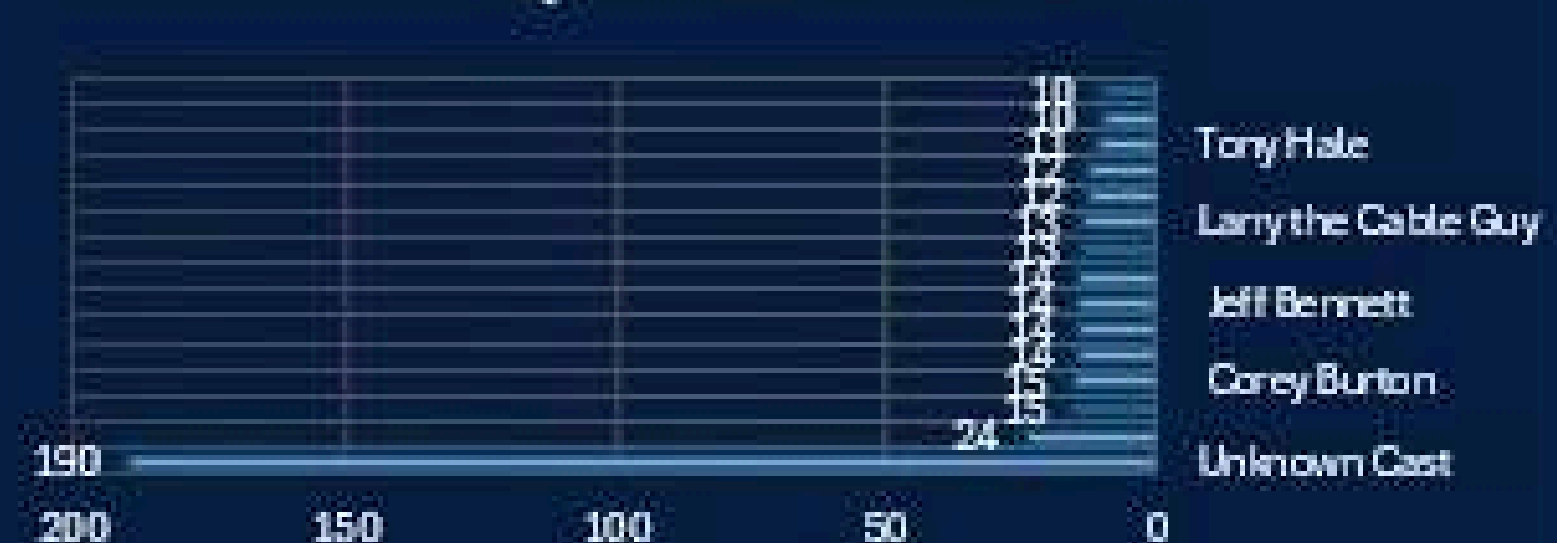
Bill Spee...

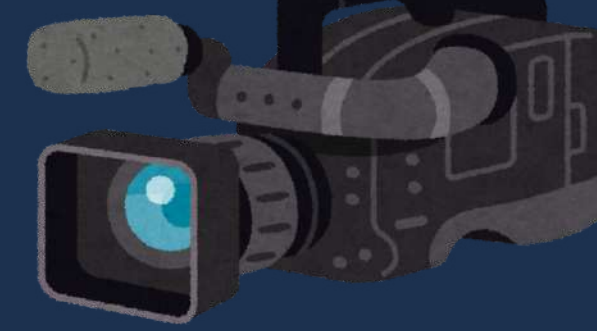
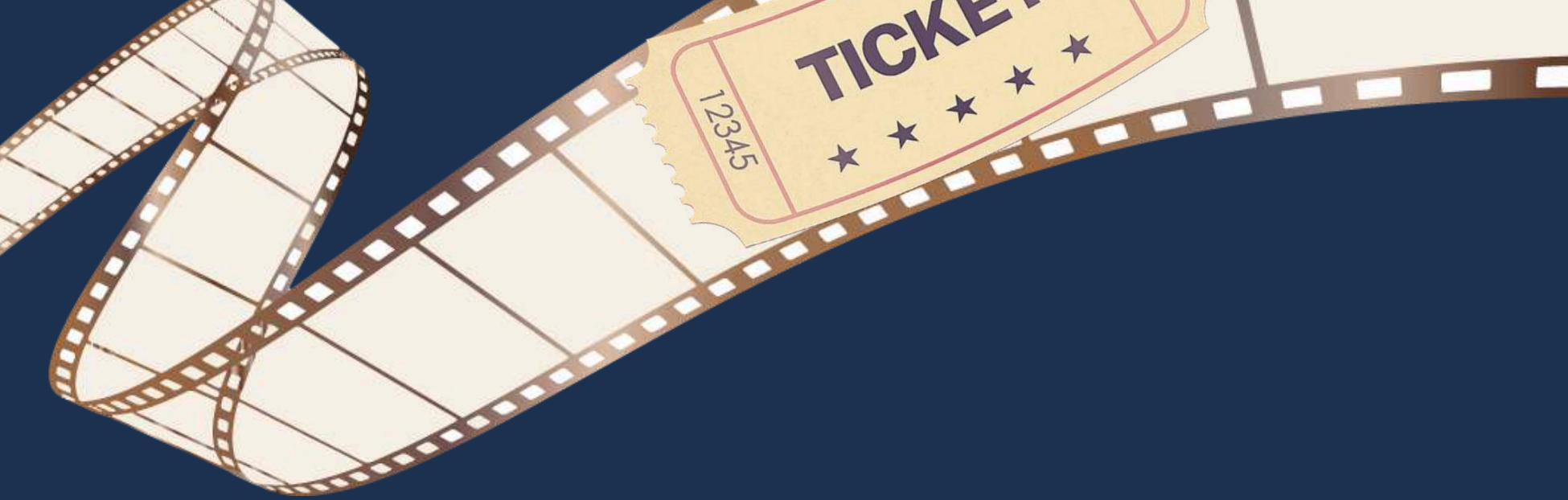
Bobs Ga...

Unknown vs Known Country Data



Top 15 Cast Members





POWER BI

Power BI





POWER BI DASHBOARD 1: RATINGS ANALYSIS



Visual Focus

Interactive breakdown of maturity ratings across Movies and TV Shows with dynamic cross-filtering.

Key Insight

The dashboard provides clear visibility into how content genres correspond to different maturity ratings, including patterns within Animation content.





DASHBOARD



POWER BI DASHBOARD 2: TIME TRENDS & KPIS



Visual Focus

Time-series analysis tracking catalog additions by year and month, supported by core KPI cards.

Key Insight

The dashboard highlights changes in catalog additions over time and provides a clear view of overall content growth.





DASHBOARD



POWER BI DASHBOARD 3: GENRES & DURATION



Visual Focus

Interactive comparison of primary genres, movie runtimes, and TV show seasons.

Key Insight

The dashboard highlights differences in content volume and duration across primary genres and TV show seasons.





RECOMMENDATIONS & STRATEGIC ACTIONS

Data Quality

- Improve Metadata Quality
- Address missing values in countries and directors at the source to enable more accurate geographical and director-level analysis.
 - ~15% Missing Countries
 - ~32% Missing Directors

Geographic Diversification

- Expand Geographic Diversity
- Broaden content sources beyond the U.S. dominant share by investing more in regions such as the UK and Canada to reach a wider global audience. ~69% U.S. Content



RECOMMENDATIONS & STRATEGIC ACTIONS

Fresh Content

- Increase Fresh Content
 - Reduce the gap between content release and platform addition by increasing fresh and day-and-date releases.
- ~ 16.6 Years Average Release-to-Addition Gap

Release Schedule

- Optimize Release Timing
 - Avoid excessive concentration of content additions in specific months and weekdays by creating a more balanced release schedule.
- ~ 56% in November ~ 89% on Tuesdays & Fridays



RECOMMENDATIONS & STRATEGIC ACTIONS



1. TV Shows Expansion
Expand TV Show Content
Balance the movie-heavy catalog by investing in more serial TV shows to support long-term user engagement.

- Movie : TV Show = 2.66 : 1

2. Demographic Expansion
Broaden Audience Segments
Increase content targeting teens and adults alongside the existing family-oriented catalog.

- PG-13 + TV-14 $\approx$ 10%
 - Family / Animation Focus $\approx$ 62.6%
- 





RECOMMENDATIONS & STRATEGIC ACTIONS

3. Runtime Strategy

Optimize Content Runtime

Use runtime patterns to inform content planning, while maintaining a balance between shorter family content and longer adult-oriented productions.

- PG-13 Movies → ~128 min
- Family Content → ~71.9 min



THANK YOU